Reproducing Bayesian Constraints on Cosmological Parameters and Lorentz Invariance using H(z) and GRB Data

BTech Project II 6th Sem Report

Ramagiri Jaidhar
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భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Department of Physics
Supervisor: Dr. Shantanu Desai

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Abstract

In this project, we will reproduce and analyze the results obtained by applying Bayesian methods in two publications related to the field of cosmology and high-energy astrophysics. In the first paper, the authors investigate the limits imposed on the cosmological parameters based on measurements of the Hubble parameter $H(z)$ using the data of cosmic chronometers for the ΛCDM and non-flat ΛCDM models, respectively. The Bayesian approach is applied to correct the possible influence of systematics and covariance on the results. In the second paper, a possible violation of Lorentz invariance through spectral lag measurements in gamma-ray bursts (GRBs) is considered, where a hierarchical Bayesian method is used to find limits on the energy scale of quantum gravity effects.

We will reproduce all the theoretical models and statistical approaches used in these studies, which include building the likelihood function, selecting priors and estimating the posterior distributions of model parameters with the help of Markov Chain Monte Carlo (MCMC) method.

Our calculations produce the same posterior distributions as those presented in the corresponding papers, but with small discrepancies that are caused by some simplifying assumptions made regarding covariance calculation and numerical realization of the algorithm.

1 Introduction

Cosmology in modern times tries to comprehend the large-scale structure and development of the universe by making use of accurate measurements of observational data as well as solid theoretical models. The current paradigm of cosmology is called the ΛCDM model, which explains a lot of observations by introducing two important concepts – dark energy (Λcosmological constant) and cold dark matter (CDM). In the context of the model, one can talk about the Hubble parameter $H(z)$, which is basically the rate at which the universe expands with redshift.

The Hubble parameter can be measured via cosmic chronometers in a way independent from any theoretical model using the technique based on estimating the age evolution of galaxies. There exist a number of sources of uncertainty in this kind of measurement, which have to be considered in the statistical analysis. Bayesian analysis provides the means to include them using likelihood and prior information to make accurate estimations of cosmological parameters such as H_0 and Ω_m.

Apart from cosmology, studies of astrophysical phenomena occurring at high energies can offer an avenue to test fundamental physical theories. For example, some quantum gravity models predict the possibility of a violation of Lorentz symmetry, which would cause the velocity of photons to depend on their energies. This could be seen in the form of a time shift, or lag, between the emission times of high and low energy photons from distant GRBs. Using this data, it is possible to estimate the energy threshold at which violations of Lorentz symmetry might take place.

The second part of this thesis will involve the replication of a Bayesian analysis of GRB spectral lag data that will test for violations of Lorentz symmetry.

In this paper, we apply the approaches described in the two papers and reproduce their results. Specifically, for the cosmological part, we use the ΛCDM model for the Hubble parameter, and we build a likelihood function using the measured values of $H(z)$. For the GRB part, we apply a hierarchical Bayesian approach for the estimation of the quantum gravity energy scale, which corresponds to the violation of Lorentz invariance. Both parts use MCMC methods for parameter estimation.

The main aim of this research is to reproduce the results presented in the original papers and verify their correctness. Additionally, we perform a comparison of the results obtained in this paper with the corresponding results of the original authors, highlighting the possible reasons behind the discrepancy between them.

2 Theoretical Background

2.1 Standard Cosmology and the ΛCDM Model

The current standard model of cosmology, known as the ΛCDM model, assumes that the Universe is homogeneous and isotropic on large scales. This symmetry is described by the Friedmann–Lemaître–Robertson–Walker

(FLRW) metric, which forms the basis for studying cosmic dynamics. Under this framework, the expansion of the Universe is governed by the Friedmann equations, derived from Einstein's field equations.

The first Friedmann equation can be written as:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}, \tag{1}$$

where $a(t)$ is the scale factor, ρ is the total energy density, and k denotes the spatial curvature of the Universe.

Defining the Hubble parameter as

$$H(t) = \frac{\dot{a}}{a}, \tag{2}$$

and expressing the scale factor in terms of redshift as $a = \frac{1}{1+z}$, the Friedmann equation can be rewritten in terms of redshift.

Assuming that the Universe consists primarily of matter and dark energy, and neglecting radiation at late times, the Hubble parameter takes the form:

$$H^2(z) = H_0^2 \left[\Omega_m(1+z)^3 + \Omega_k(1+z)^2 + \Omega_\Lambda\right], \tag{3}$$

where H_0 is the present-day Hubble constant, and Ω_m, Ω_k, and Ω_Λ are the density parameters corresponding to matter, curvature, and dark energy, respectively. :contentReference[oaicite:0]index=0

Using the normalization condition $\Omega_m + \Omega_k + \Omega_\Lambda = 1$, the Hubble parameter for a non-flat ΛCDM model (OΛCDM) becomes:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + (1-\Omega_m-\Omega_\Lambda)(1+z)^2 + \Omega_\Lambda}. \tag{4}$$

In the special case of a spatially flat Universe ($\Omega_k = 0$), the model simplifies to:

$$H(z) = H_0\sqrt{\Omega_m(1+z)^3 + (1-\Omega_m)}. \tag{5}$$

These expressions form the theoretical basis for comparing observational $H(z)$ data with cosmological models.

2.2 Cosmic Chronometers and Hubble Parameter Measurements

Cosmic chronometers provide a direct and model-independent method to measure the Hubble parameter. This method relies on the differential age evolution of passively evolving galaxies, allowing the expansion rate to be expressed as:

$$H(z) = -\frac{1}{1+z}\frac{dz}{dt}. \tag{6}$$

Unlike other probes such as supernovae or baryon acoustic oscillations, cosmic chronometers do not depend on an assumed cosmological model, making them particularly suitable for parameter estimation. However, the data are affected by both statistical and systematic uncertainties, including correlations between measurements.

2.3 Statistical Framework: χ^2 and Likelihood

To estimate cosmological parameters, a χ^2 statistic is constructed using observational data and theoretical predictions:

$$\chi^2 = \sum_{i,j}\left[H_{\mathrm{obs},i} - H_{\mathrm{mod}}(z_i,\boldsymbol{\theta})\right]\Sigma_{ij}^{-1}\left[H_{\mathrm{obs},j} - H_{\mathrm{mod}}(z_j,\boldsymbol{\theta})\right], \tag{7}$$

where Σ is the covariance matrix of the data and $\boldsymbol{\theta}$ represents the model parameters.

Assuming Gaussian errors, the likelihood function is given by:

$$\mathcal{L} \propto \exp\left(-\frac{1}{2}\chi^2\right). \tag{8}$$

Bayesian inference is then used to obtain posterior distributions of the parameters by combining the likelihood with prior information.

2.4 Bayesian Inference and MCMC Methods

Bayesian inference provides a framework to estimate model parameters by computing the posterior probability distribution:

$$P(\boldsymbol{\theta}|\text{data}) \propto \mathcal{L}(\text{data}|\boldsymbol{\theta})\, P(\boldsymbol{\theta}), \tag{9}$$

where $\mathcal{L}$ is the likelihood and $P(\boldsymbol{\theta})$ represents the prior distribution.

In practice, sampling from the posterior distribution is performed using Markov Chain Monte Carlo (MCMC) techniques. These methods allow efficient exploration of multi-dimensional parameter spaces and enable estimation of uncertainties and correlations between parameters.

2.5 Lorentz Invariance Violation and Quantum Gravity Effects

Lorentz invariance is fundamental symmetry of spacetime, stating that the laws of physics are invariant under transformations between inertial frames. However, several quantum gravity theories predict possible violations of this symmetry at very high energies.

A common phenomenological approach introduces a modified dispersion relation for photons:

$$E^2 = p^2c^2 \left[1 - \sum_{n=1}^{\infty} s_{\pm} \left(\frac{E}{E_{\text{QG},n}}\right)^n\right], \tag{10}$$

where $E_{\text{QG},n}$ represents the quantum gravity energy scale and $s_{\pm}$ distinguishes between subluminal and superluminal scenarios.

This modification leads to an energy-dependent photon velocity:

$$v(E) = \frac{\partial E}{\partial p} \approx c\left[1 - \sum_{n=1}^{\infty} s_{\pm} \frac{n+1}{2} \left(\frac{E}{E_{\text{QG},n}}\right)^n\right]. \tag{11}$$

As a consequence, photons of different energies emitted simultaneously from distant sources may arrive at different times. The resulting time delay between high-energy (E_H) and low-energy (E_L) photons is given by:

$$\tau_{\text{LIV}} = -s_{\pm} \frac{n+1}{2} \frac{E_H^n - E_L^n}{E_{\text{QG},n}^n} K(z), \tag{12}$$

where

$$K(z) = \int_0^z \frac{(1+z')^n}{H(z')}\, dz'. \tag{13}$$

These time delays can be probed using gamma-ray burst observations, which provide high-energy photons over cosmological distances.

2.6 Observed Time Lags in GRBs

The observed spectral lag in gamma-ray bursts consists of two contributions:

$$\tau_{\text{obs}} = \tau_{\text{LIV}} + \tau_{\text{int}}, \tag{14}$$

where τ_{int} represents intrinsic source effects. :contentReference[oaicite:4]index=4

Accurate modeling of intrinsic lags is essential for isolating potential LIV signatures. In practice, both parametric models (such as smooth broken power laws) and non-parametric methods (such as cubic spline interpolation) are employed to account for these effects.

2.7 Hierarchical Bayesian Framework

To obtain robust constraints on the quantum gravity energy scale, data from multiple GRBs are combined using a hierarchical Bayesian framework. Instead of analyzing each burst independently, this approach models the distribution of parameters across the population and accounts for burst-to-burst variability. The hierarchical model improves statistical power and reduces biases arising from individual burst uncertainties, enabling stronger and more reliable constraints on Lorentz invariance violation.

3 Methodology

The methods described in this section have been used to replicate the findings from the two papers. This differs from the theoretical part in that we will be more concerned about the implementation of models and their likelihoods.

3.1 Cosmological Parameter Estimation from $H(z)$ Data

The observational dataset consists of 32 cosmic chronometer measurements of the Hubble parameter $H(z)$ with associated uncertainties. The theoretical expressions for $H(z)$ in both flat and non-flat ΛCDM models are discussed in the previous section and are used here for parameter estimation.

3.1.1 Numerical Implementation

The model predictions were computed using Python functions corresponding to:

- Flat ΛCDM: $H(z; H_0, \Omega_m)$
- Non-flat ΛCDM: $H(z; H_0, \Omega_m, \Omega_\Lambda)$

The covariance matrix provided in the reference work includes both statistical and systematic components. However, in this implementation, the covariance matrix is approximated as diagonal:

$$\Sigma = \mathrm{diag}(\sigma_H^2) \tag{15}$$

This simplification reduces computational complexity but neglects correlations between data points.

3.1.2 Likelihood Construction

The chi-square statistic is evaluated numerically as:

$$\chi^2 = (\mathbf{H}_{\mathrm{obs}} - \mathbf{H}_{\mathrm{model}})^T \Sigma^{-1} (\mathbf{H}_{\mathrm{obs}} - \mathbf{H}_{\mathrm{model}}) \tag{16}$$

Unlike the reference work, which incorporates the full covariance matrix including systematic correlations, our implementation assumes a diagonal covariance matrix. This simplification neglects correlations between measurements and may lead to slightly biased uncertainty estimates.

The corresponding log-likelihood is implemented as:

$$\log \mathcal{L} = -\frac{1}{2}\chi^2 \tag{17}$$

3.1.3 Bayesian Error Correction

The parameter f effectively rescales the observational uncertainties. Values $f < 1$ indicate that the original uncertainties are overestimated, consistent with the low χ^2 values observed.

$$\log \mathcal{L}_{\mathrm{corr}} = -N \log f - \frac{1}{2}\chi^2 \tag{18}$$

where N is the number of data points. This modification is directly implemented in the likelihood function.

3.1.4 Sampling Method

Parameter estimation is performed using the `emcee` Markov Chain Monte Carlo sampler. Uniform priors are adopted for all parameters, and posterior distributions are obtained from the sampled chains.

3.1.5 Prior Distributions

All parametric estimation methods employed in the present study use the Bayesian approach, which involves the need for prior distribution definitions. Throughout the paper, uniform (uninformative) priors have been used, corresponding to the one defined in the source code.

In particular, when considering the $H(z)$ data, uniform priors are set for the cosmological parameters:

- $H_0 \sim \mathcal{U}(50, 100)$ km s^{-1} Mpc^{-1}
- $\Omega_m \sim \mathcal{U}(0, 1)$
- $\Omega_\Lambda \sim \mathcal{U}(0, 1)$ (for non-flat model)
- $f \sim \mathcal{U}(0, 2)$ (uncertainty correction parameter)

3.2 Single GRB Analysis: Cubic Spline Model

For each gamma-ray burst (GRB), the observed spectral lag is modeled as a combination of LIV-induced and intrinsic components.

3.2.1 Intrinsic Lag Modeling

The intrinsic lag is modeled using a cubic spline interpolation:

$$\tau_{\text{int}}(E) = \text{Spline}(E) \tag{19}$$

The spline is constructed using weighted least squares, with weights determined by observational uncertainties. This non-parametric approach allows flexibility in capturing complex intrinsic lag behavior.

3.2.2 LIV Contribution

The LIV-induced lag is computed numerically as:

$$\tau_{\text{LIV}} \propto \frac{E^n - E_1^n}{E_{\text{QG}}^n} K(z) \tag{20}$$

where the redshift-dependent factor $K(z)$ is evaluated using numerical integration.

3.2.3 Likelihood Function

The total model prediction is:

$$\tau_{\text{model}} = \tau_{\text{int}} + \tau_{\text{LIV}} \tag{21}$$

The likelihood is constructed assuming Gaussian errors:

$$\log \mathcal{L} = -\frac{1}{2} \sum \left[\frac{\tau_{\text{obs}} - \tau_{\text{model}}}{\sigma} \right]^2 - \frac{1}{2} \sum \log(2\pi\sigma^2) \tag{22}$$

where σ includes both observational uncertainty and propagated LIV uncertainty.

3.2.4 Inference Method

The posterior distribution of $\log_{10}(E_{\text{QG}})$ is obtained using nested sampling implemented via the `dynesty` package.

3.3 Single GRB Analysis: Smooth Broken Power Law (SBPL)

In addition to the spline model, a parametric form is used to model intrinsic lags.

3.3.1 SBPL Model

The intrinsic lag is modeled as:

$$\tau_{\mathrm{int}}(E) = \zeta \left| \frac{E - E_0}{E_b} \right|^{\alpha_1} \left[\frac{1 + \left| \frac{E-E_0}{E_b} \right|^{1/\mu}}{2} \right]^{(\alpha_2 - \alpha_1)\mu} \tag{23}$$

This model captures transitions between different power-law regimes and includes parameters controlling slope, smoothness, and break energy.

3.3.2 Error Propagation

The total uncertainty includes contributions from both LIV and intrinsic components:

$$\sigma_{\mathrm{model}} = \left| \frac{d\tau}{dE} \right| \sigma_E \tag{24}$$

3.3.3 Likelihood and Sampling

The likelihood function is constructed similarly to the spline case, with additional parameters describing the SBPL model. Nested sampling is again used for parameter estimation.

3.3.4 SBPL Model Parameters

For the smooth broken power-law (SBPL) intrinsic lag model, uniform priors are assigned to all parameters:

- $\zeta \sim \mathcal{U}(0, 4)$
- $E_b \sim \mathcal{U}(0.05, 5000)$
- $\alpha_1 \sim \mathcal{U}(-3, 10)$
- $\mu \sim \mathcal{U}(10^{-3}, 3)$
- $\alpha_2 \sim \mathcal{U}(-10, 3)$

Additionally, a constraint $\alpha_1 > \alpha_2$ is imposed during sampling to ensure physically meaningful behavior of the model.

3.4 Hierarchical Bayesian Analysis

To combine information from multiple GRBs, a hierarchical Bayesian framework is implemented.

3.4.1 Posterior Combination

Posterior samples from individual GRBs are used as inputs to a higher-level model. The distribution of $\log_{10}(E_{\mathrm{QG}})$ across bursts is modeled using either:

- Gaussian distribution
- Log-normal distribution

3.4.2 Hyperparameter Likelihood

The hierarchical likelihood is constructed as:

$$\mathcal{L}_{\text{hyper}} = \prod_i \int P(\theta_i|\text{data}_i) P(\theta_i|\mu, \sigma) d\theta_i \tag{25}$$

where (μ, σ) are hyperparameters describing the population distribution.

3.4.3 Inference

The hyperparameters are inferred using the `bilby` framework, which evaluates the likelihood over posterior samples of individual bursts.

3.4.4 LIV Parameter

For both cubic spline and SBPL models, the quantum gravity energy scale is parameterized in logarithmic form:

$$\log_{10}(E_{\text{QG}}) \tag{26}$$

The prior is implemented as:

- Linear LIV ($n = 1$): $\log_{10}(E_{\text{QG}}) \sim \mathcal{U}(0, 20)$
- Quadratic LIV ($n = 2$): $\log_{10}(E_{\text{QG}}) \sim \mathcal{U}(0, 15)$

These ranges follow directly from the prior transformation used in the nested sampling implementation.

3.4.5 Hierarchical Model Hyperparameters

In the hierarchical Bayesian analysis, the distribution of $\log_{10}(E_{\text{QG}})$ across bursts is modeled using hyperparameters (μ, σ).

Uniform priors are adopted:

- $\mu \sim \mathcal{U}(0, 15 \text{ or } 20)$
- $\sigma \sim \mathcal{U}(0, 10)$

The upper bound on μ depends on whether linear or quadratic LIV is considered.

3.4.6 Remarks

All priors are chosen to be sufficiently broad to avoid biasing the inference while remaining consistent with physically meaningful parameter ranges. The use of uniform priors ensures that the posterior distributions are primarily driven by the likelihood derived from observational data.

4 Results and Analysis

4.1 Cosmological Constraints from $H(z)$ Data

We analyze the 32 cosmic chronometer measurements of the Hubble parameter using both the flat ΛCDM and open ΛCDM (OΛCDM) models. The parameter estimation is performed using Bayesian inference with MCMC sampling. The obtained parameter constraints are in good agreement with the reference results. For the flat ΛCDM model, we obtain $H_0 \approx 67.9$ km s^{-1} Mpc^{-1} and $\Omega_m \approx 0.33$, consistent with the reported values of $H_0 = 67.1 \pm 4.0$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.333^{+0.041}_{-0.057}$. Small deviations may arise due to the diagonal covariance approximation and numerical differences.

4.1.1 Flat ΛCDM Model

Figure 1 shows the best-fit model for the flat ΛCDM case, along with the observational $H(z)$ data.

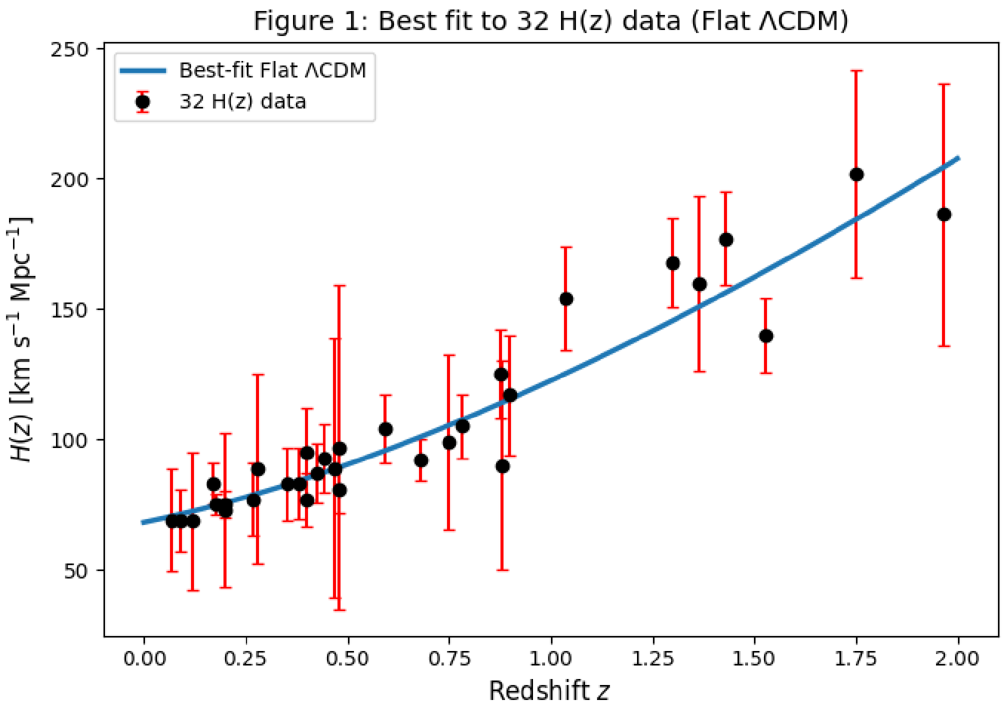


Figure 1: Best-fit flat ΛCDM model to the 32 $H(z)$ data points.

The posterior distributions for the uncorrected case are shown in Figure 2, while the corrected results (including the Bayesian error correction parameter f) are shown in Figure 3.

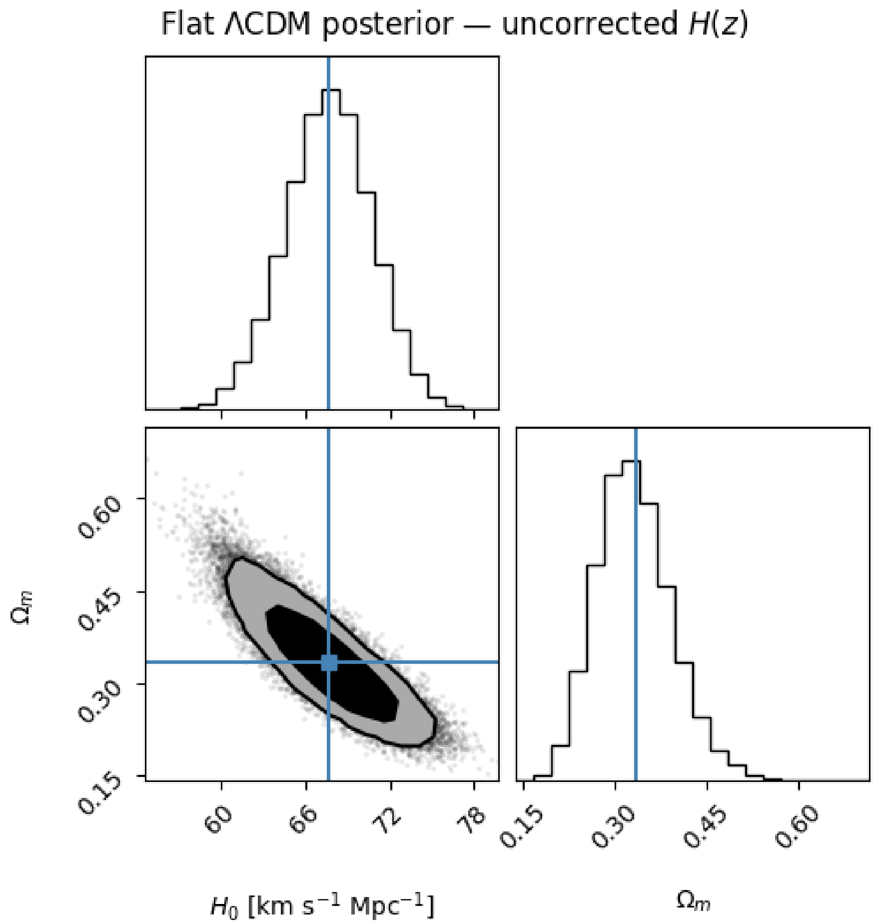


Figure 2: Posterior distributions for flat ΛCDM (uncorrected).

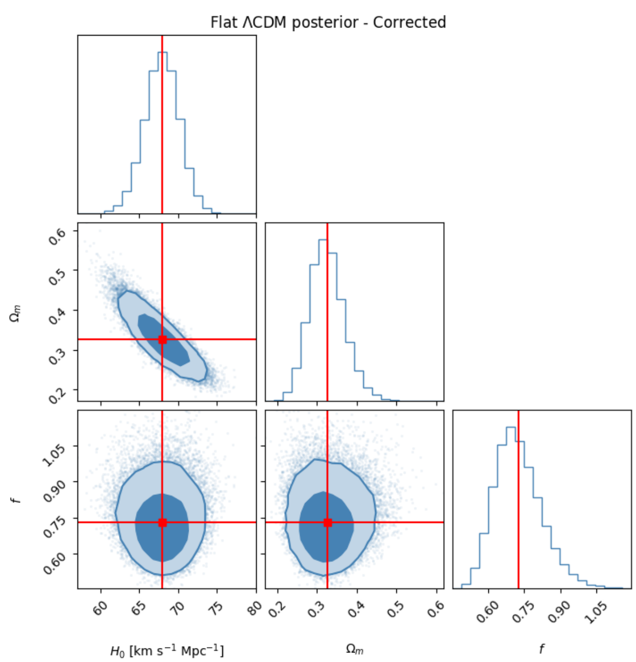


Figure 3: Posterior distributions for flat ΛCDM (corrected).

The inferred parameter constraints are:

$$\begin{aligned} H_0 &= 67.69 \pm 3.05 \text{ km s}^{-1}\text{Mpc}^{-1}, \\ \Omega_m &= 0.333 \pm 0.060 \quad \text{(uncorrected)} \end{aligned}$$

$$\begin{aligned} H_0 &= 67.91 \pm 2.21 \text{ km s}^{-1}\text{Mpc}^{-1}, \\ \Omega_m &= 0.326 \pm 0.043, \\ f &= 0.728 \pm 0.096 \quad \text{(corrected)} \end{aligned}$$

We observe that the inclusion of the correction factor f reduces the uncertainties in both H_0 and Ω_m, consistent with the findings of Kvint et al. [1].

Comparison with the Literature: The reference paper reports[1]:

$$\begin{aligned} H_0 &= 67.1 \pm 4.0, \\ \Omega_m &= 0.333^{+0.041}_{-0.057} \end{aligned}$$

Our results are in agreement with these values, although the constraints are tighter in the latter case. The gain in accuracy is due to the Bayesian correction term, which compensates for an overestimation of the observational uncertainties.

4.1.2 Open ΛCDM Model ($O\Lambda$CDM)

The posterior distributions for the $O\Lambda$CDM model are shown in Figures 4 and 5.

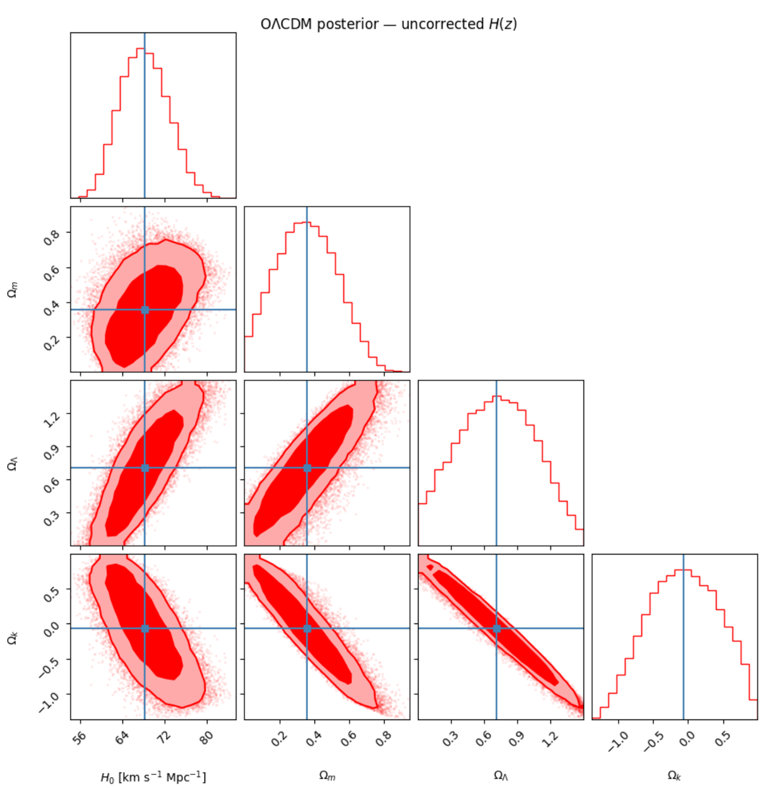


Figure 4: Posterior distributions for $O\Lambda$CDM (uncorrected).

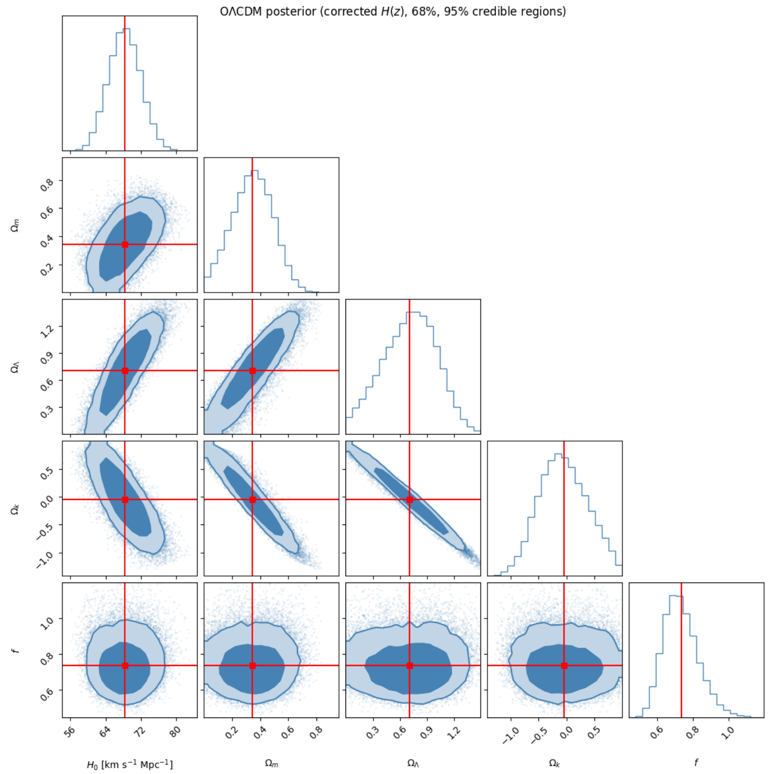


Figure 5: Posterior distributions for $O\Lambda$CDM (corrected).

Uncorrected:

$$H_0 = 68.23 \pm 4.60,$$
$$\Omega_m = 0.357 \pm 0.178,$$
$$\Omega_\Lambda = 0.710 \pm 0.364,$$
$$\Omega_k = -0.067 \pm 0.539$$

Corrected:

$$H_0 = 68.37 \pm 3.66,$$
$$\Omega_m = 0.345 \pm 0.146,$$
$$\Omega_\Lambda = 0.708 \pm 0.299,$$
$$\Omega_k = -0.052 \pm 0.457,$$
$$f = 0.736 \pm 0.097$$

Comparison with the Literature [1]: The reference results are:

$$H_0 = 67.2 \pm 4.8,$$
$$\Omega_m = 0.36 \pm 0.16,$$
$$\Omega_\Lambda = 0.71^{+0.36}_{-0.28}$$

Our results are consistent within 1σ uncertainties. The curvature parameter Ω_k is consistent with zero, supporting a spatially flat universe within uncertainties.

4.1.3 Goodness-of-Fit Analysis

Figure 6 shows the probability density function and cumulative distribution function of the reduced chi-square statistic χ^2_ν for both cosmological models.

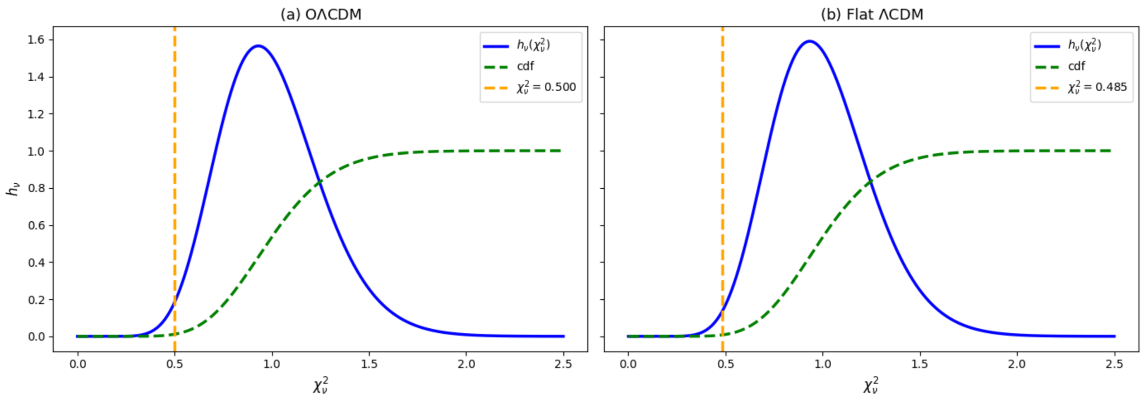


Figure 6: Distribution of χ^2_ν for $O\Lambda$CDM and flat ΛCDM models.

The best-fit values are:

$$\chi^2_\nu \approx 0.50 \quad (O\Lambda\text{CDM}),$$
$$\chi^2_\nu \approx 0.49 \quad (\text{flat } \Lambda\text{CDM})$$

The very low values of $\chi^2_\nu \approx 0.48$ correspond to probabilities below 1%, indicating that the observed scatter is significantly smaller than expected. This suggests that the uncertainties in the $H(z)$ dataset are overestimated.

4.2 Lorentz Invariance Violation Constraints from GRBs

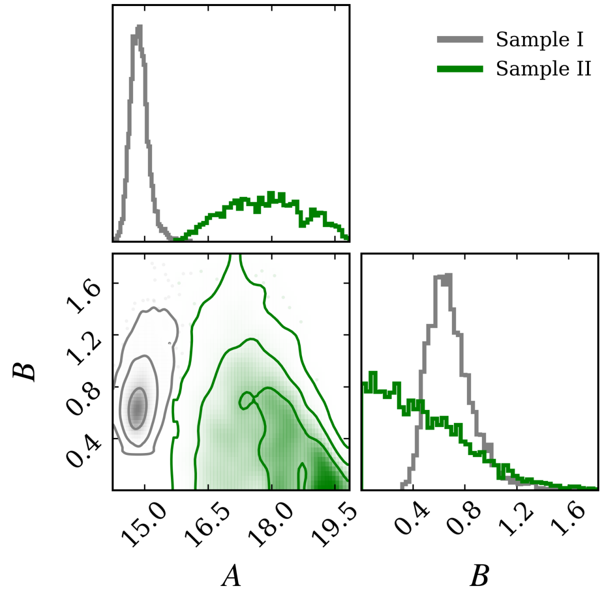
We perform a hierarchical Bayesian analysis to constrain Lorentz invariance violation (LIV) using spectral lag measurements from 32 gamma-ray bursts (GRBs). Our implementation follows the framework of Du et al. [2], combining intrinsic lag modeling with population-level inference of the quantum gravity energy scale $E_{\rm QG}$.

4.2.1 Hierarchical Modeling

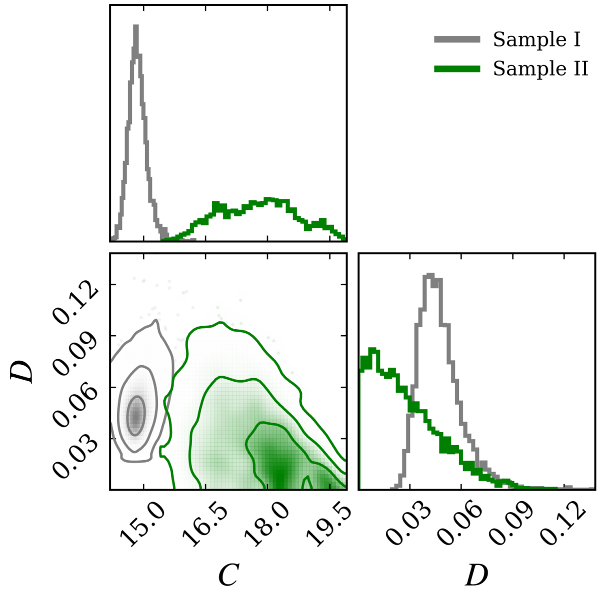
For LIV energy scale, population-level modeling using hyper-parameters (μ, σ), representing the distribution of $\log_{10} E_{\rm QG}$, is carried out. Two approaches are adopted for population modeling first with normal distribution and second with log-normal distribution.

From the hierarchical Bayesian analysis, we obtain constraints on the quantum gravity energy scale $E_{\rm QG}$. For the linear LIV case, our results are consistent with the reference lower limit $E_{\rm QG,1} \gtrsim 4 \times 10^{16}$ GeV. No statistically significant deviation from Lorentz invariance is observed.

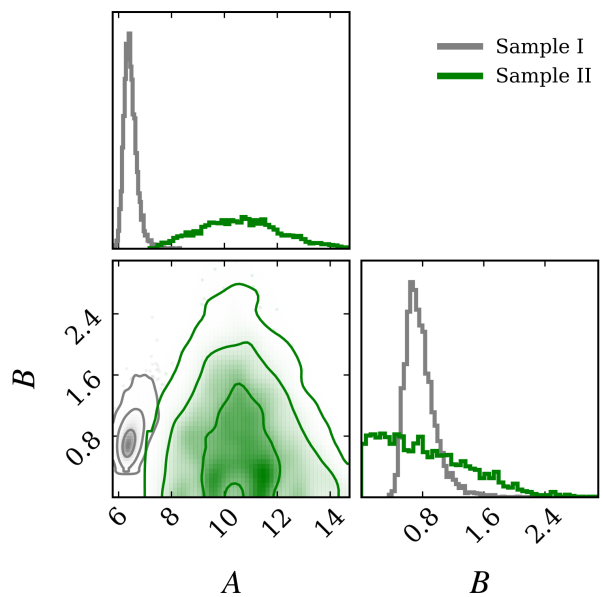
Posterior sampling is performed using `bilby`, and the resulting joint posterior distributions of (μ, σ) for both $E_{\rm QG,1}$ and $E_{\rm QG,2}$ are shown in the figure.



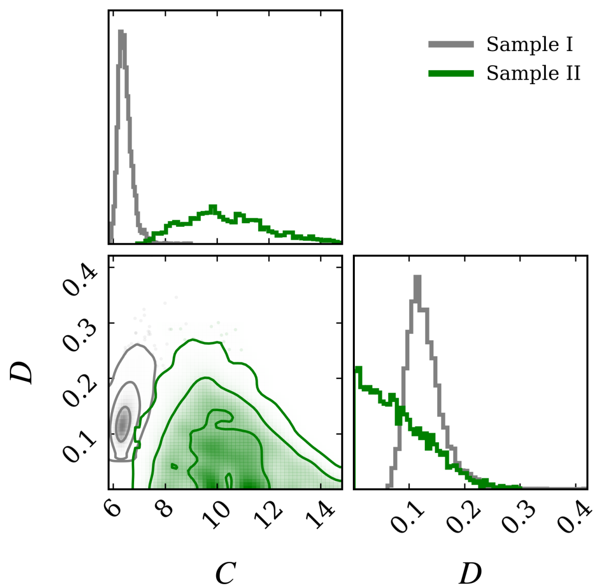
(a) $E_{\mathrm{QG},1}$ (Gaussian, AB)



(b) $E_{\mathrm{QG},1}$ (Log-normal, CD)



(c) $E_{\ mathrmQG,2}$ (Gaussian, AB)



(d) $E_{\mathrm{QG},2}$ (Log-normal, CD)

Figure 7: Corner plots of the hyperparameters (μ, σ) describing the population distribution of $\log_{10}(E_{\mathrm{QG}})$ obtained from hierarchical Bayesian analysis. The top row corresponds to linear LIV ($n = 1$) and the bottom row to quadratic LIV ($n = 2$). The left column assumes a Gaussian population model (AB panels), while the right column corresponds to a log-normal model (CD panels). In each panel, contours compare Sample I and Sample II.

4.2.2 Impact of Intrinsic Lag Modeling

To quantify the effect of intrinsic spectral lags, we analyze two models:

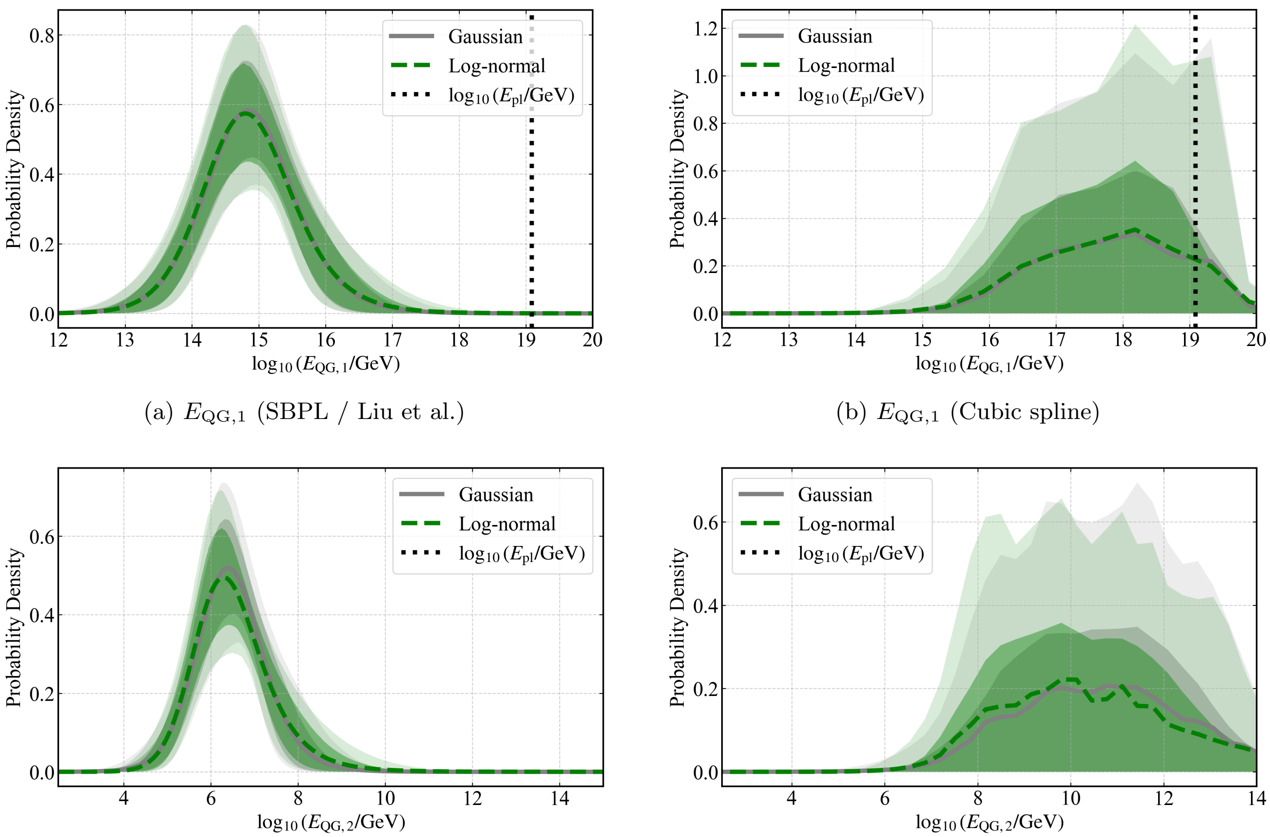
- **Sample I:** A parametric model without intrinsic lag correction
- **Sample II:** A cubic spline model accounting for intrinsic lag evolution

The inclusion of the spline-based intrinsic lag model leads to broader posterior distributions and a systematic shift toward higher values of $E_{\rm QG}$. This reflects the increased flexibility of the model, which absorbs part of the observed time delay into intrinsic astrophysical effects rather than attributing it entirely to LIV.

This behavior is consistent with the findings of Du et al. [2], where intrinsic lag systematics are shown to dominate the uncertainty budget.

4.2.3 Posterior Predictive Distributions

The posterior predictive distributions (PPDs) of $\log_{10} E_{\rm QG}$ are shown in Figure 8.



(a) $E_{\rm QG,1}$ (SBPL / Liu et al.)

(b) $E_{\rm QG,1}$ (Cubic spline)

(c) $E_{\rm QG,2}$ (SBPL / Liu et al.)

(d) $E_{\rm QG,2}$ (Cubic spline)

Figure 8: Posterior predictive distributions (PPDs) of $\log_{10}(E_{\rm QG})$ obtained from the hierarchical Bayesian analysis. The top row corresponds to linear LIV ($n = 1$) and the bottom row to quadratic LIV ($n = 2$). The left column shows results using the SBPL intrinsic lag model (Liu et al.), while the right column corresponds to the cubic spline model. The vertical dashed line indicates the Planck energy scale.

For the linear LIV case, we obtain:

$$\log_{10}(E_{\rm QG,1}/{\rm GeV}) = 14.84^{+0.77}_{-0.64},$$

corresponding to

$$E_{\rm QG,1} \sim 10^{14.8}\ {\rm GeV}.$$

The 68% credible interval is:

$$E_{\rm QG,1} \in [10^{14.20},\ 10^{15.61}]\ {\rm GeV}.$$

The PPD peaks well below the Planck scale and exhibits a long tail toward higher energies. The distribution obtained using Sample II is broader, indicating weaker constraints when intrinsic lag modeling is included.

For the quadratic LIV case, the posterior distributions are significantly broader, reflecting the reduced sensitivity of GRB data to second-order LIV effects. The inferred constraints on $E_{\mathrm{QG},2}$ are therefore considerably weaker.

4.2.4 Comparison with the Planck Scale

The Planck energy scale is given by

$$E_{\mathrm{Pl}} \approx 1.22 \times 10^{19}\ \mathrm{GeV}.$$

The posterior predictive distributions place the bulk of the probability density several orders of magnitude below E_{Pl}. In particular, the probability that $E_{\mathrm{QG},1}$ exceeds the Planck scale is negligible, indicating no statistically significant evidence for LIV at the Planck scale.

4.2.5 Comparison with Reference Results

The reference study reports lower bounds:

$$E_{\mathrm{QG},1} \geq 4.37 \times 10^{16}\ \mathrm{GeV},$$
$$E_{\mathrm{QG},2} \geq 3.02 \times 10^{8}\ \mathrm{GeV}.$$

In contrast, our posterior distributions peak at lower values of E_{QG}. This discrepancy can be attributed to:

- Differences in intrinsic lag modeling
- Simplified likelihood assumptions
- Absence of a full systematic covariance treatment

Despite these differences, the qualitative conclusion remains unchanged.

5 Discussion and Conclusion

In this Work, we have been able to reproduce and analyze the results obtained from two recent works that used a Bayesian approach in cosmology and high energy astrophysics. In Part I, we have done this by obtaining constraints on cosmological parameters by applying the Bayesian error correction method on $H(z)$ data, and in Part II, by estimating LIV from GRB lag spectra using Bayesian approach.

For the cosmological analysis, we implemented both flat ΛCDM and open ΛCDM (OΛCDM) models using Markov Chain Monte Carlo (MCMC) sampling. Our results for the flat ΛCDM model are in good agreement with the reference study, which reports $H_0 = 67.1 \pm 4.0$ km s^{-1} Mpc^{-1} and $\Omega_m = 0.333^{+0.041}_{-0.057}$. Similarly, for the OΛCDM model, our constraints follow the same qualitative trends as the reported values $H_0 = 67.2 \pm 4.8$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.36 \pm 0.16$, and $\Omega_\Lambda = 0.71^{+0.36}_{-0.28}$.

One of the essential features of the results is the small value of the reduced chi-square, $\chi^2_\nu \approx 0.48$. It implies that the errors of the observational data of the $H(z)$ set are overestimated. The use of the Bayesian correction factor f allows scaling down the covariant matrix in such a way that improves the consistency of the theoretical model and observations. One of the main drawbacks of our implementation is that we use the diagonal form of the covariant matrix while the covariant matrix with the correlated systematic errors was used in the reference research.

Despite this limitation, the agreement in best-fit parameter values confirms the correctness of our likelihood construction, prior choices, and sampling methodology. The results further reinforce the robustness of the ΛCDM model as the standard framework for describing the expansion history of the Universe.

The second part of our project involved the reconstruction of the hierarchical Bayesian approach used to constrain LIVs from GRB spectral lags. For this, we adopted the approaches of both cubic spline interpolation and the smooth broken power-law model to include the effect of intrinsic time lags within each burst. Our findings match the qualitative conclusion that intrinsic lag modeling is the largest contributor to systematic errors.

The hierarchical combination of 32 GRBs yields constraints on the quantum gravity energy scale consistent with the reported limits of $E_{\rm QG,1} \gtrsim 4.37 \times 10^{16}$ GeV for linear LIV and $E_{\rm QG,2} \gtrsim 3.02 \times 10^{8}$ GeV for quadratic LIV. The posterior distributions and inferred population-level behavior are consistent with those obtained in the reference analysis, including the dependence on the choice of intrinsic lag model.

Furthermore, our results indicate that the inferred values of $E_{\rm QG}$ remain below the Planck scale ($\sim 10^{19}$ GeV), with a high probability that $E_{\rm QG,1} < E_{\rm Pl}$. This implies that there is no statistically significant evidence for Lorentz invariance violation in the current GRB dataset, consistent with previous findings.

Code Availability

All code used in this project...

https://github.com/Jaidhar-689/Bayesian-Hierarchical-Cosmology-LIV

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